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Interact With The Network Using mxpy

What is mxpy?

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mxpy is a CLI tool facilitating the interaction with the MultiversX blockchain and with Smart Contracts.

Key features:

- Create and convert wallets
- Create, sign and broadcast transactions
- Deploy, upgrade and interact with Smart Contracts

Installing and upgrading mxpy

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Install via pipx

- pipx install mxpy

Upgrade mxpy

- pipx upgrade mxpy

Creating wallets

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Wallet formats

- raw-mnemonic
- keystore-mnemonic
- keystore-secret-key
- pem

mxpy wallet new --format keystore-mnemonic --outfile keystore.json

Converting wallets

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Convert from keystore to pem file

```
mxpy wallet convert \
--in-format keystore-mnemonic --infile keystore.json \
--out-format pem --outfile wallet.pem
```

Creating and sending transactions

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Move balance transaction

```
mxpy tx new --pem wallet.pem \
--receiver erd1qyu5wthldzr8wx5c9ucg8kjagg0jfs53s8nr3zpz3hypefsdd8ssycr6th \
--value 10000000000000000000 \
--proxy https://devnet-gateway.multiversx.com \
--send
```

Creating and sending transactions

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ESDT Transfers

```
mxpy tx new --pem wallet.pem \
--receiver erd1qyu5wthldzr8wx5c9ucg8kjagg0jfs53s8nr3zpz3hypefsdd8ssycr6th \
--token-transfers <identifier> <amount> \
--proxy https://devnet-gateway.multiversx.com \
--send
```

Deploying a Smart Contract

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```
mxpy contract deploy --pem wallet.pem \
--bytecode adder/output/adder.wasm \
--arguments 0 \
--proxy https://devnet-gateway.multiversx.com \
--send
```

Calling a Smart Contract

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```
mxpy contract call <contract address> \
 --pem wallet.pem \
-- function add --arguments 7 \
--proxy https://devnet-gateway.multiversx.com \
--send
```

Querying a Smart Contract

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Read data from the contract via a view function

```
mxpy contract query <contract address> \
 -- function getSum
--proxy https://devnet-gateway.multiversx.com
```

Upgrading a Smart Contract

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```
mxpy contract upgrade <contract address> \
--bytecode new_code.wasm \
--arguments 0 \
--pem wallet.pem \
--proxy https://devnet-gateway.multiversx.com \
--send
```

Conclusion

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What have we done

Interacted with the MultiversX Blockchain

What have we learned

Create and convert wallets

Create and broadcast transactions

Deploy and interact smart contracts

Query account state

Conclusion

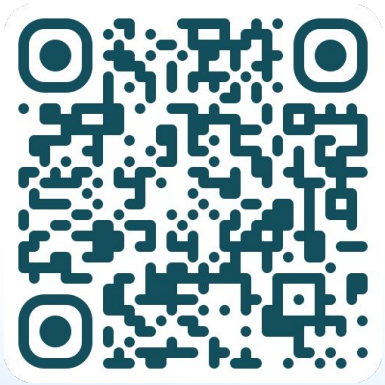
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What can we check out after this:

- Multisig Commands
- Governance Commands
- Guarded Transactions
- Localnet
- Delegation Commands

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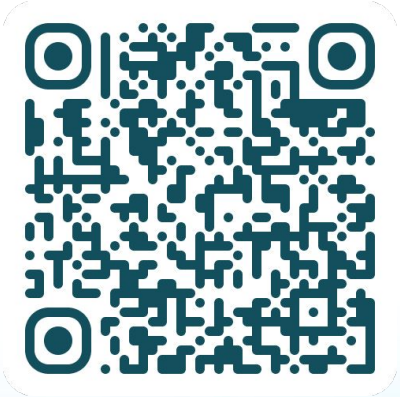
Thank you,
Have fun building!



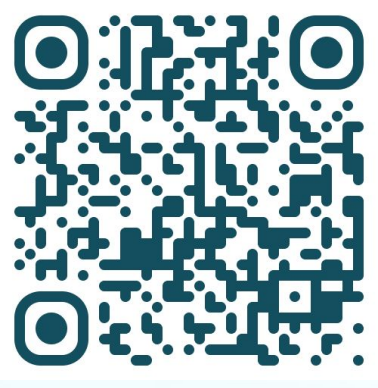
https://github.com/multiversx/mx-sdk-py-cli



https://github.com/multiversx/mx-presentations



https://docs.multiversx.com/sdk-and-tools/mxpy/mxpy-cli



https://github.com/multiversx/mx-sdk-py